MIEB: Massive Image Embedding Benchmark

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Abstract

Image and image-text embeddings underpin a wide range of vision applications, yet evaluation remains fragmented across task-specific protocols that obscure how well a model truly generalizes. We introduce MIEB, the ***Massive Image Embedding Benchmark****, a unified evaluation suite spanning 130 tasks across 8 categories and 38 languages. MIEB covers Retrieval (including interleaved, multilingual, and instruction-aware variants), Document Understanding, Linear Probing, Zero-shot Classification, Clustering, Compositionality, Vision-centric QA, and Visual STS—a novel protocol that renders text-similarity pairs as images to probe visual text understanding. All models are evaluated zero-shot (or with frozen-encoder linear probes), enabling fully reproducible comparisons via a public leaderboard built on the MTEB codebase. We additionally introduce MIEB-lite, a 51-task efficient subset that preserves near-perfect ranking correlation (Spearman $\rho = 0.992$) with an $\sim$82% runtime reduction. Evaluating 20+ models, we find that no single model dominates all categories: Voyage-multimodal-3 leads overall (mean 62.5), while CLIP/SigLIP variants excel at linear probing and zero-shot classification, and MLLM-based embeddings (E5-V, VLM2Vec) are strongest on OCR-heavy and multilingual tasks. MIEB reveals complementary strengths across model families and provides the community with the most comprehensive, reproducible image embedding evaluation to date.*

1. Introduction

Visual representations learned by image and image-text encoders are the backbone of modern vision systems, driving applications from cross-modal search [23] and document retrieval [11] to zero-shot recognition [30] and multimodal reasoning [28]. Despite this central role, evaluation of image embeddings has remained largely fragmented: models are typically assessed on isolated benchmarks—ImageNet linear probes, MSCOCO retrieval, or individual compositionality tests—making it difficult to compare models across use cases or understand where a given encoder truly excels or falls short.

This fragmentation echoes the situation that motivated the Massive Text Embedding Benchmark (MTEB) [22] for language models, which unified 56 text-embedding tasks and dramatically accelerated progress in that field. Subsequent work extended the principle to multilingual settings with MMTEB [10] and multilingual text embeddings [9]. No analogous unified benchmark exists for *visual* embeddings.

We close this gap with **MIEB**, the Massive Image Embedding Benchmark. MIEB unifies 130 tasks across 8 task categories and 38 languages, covering modalities ranging from pure image–image and image–text retrieval to interleaved and instruction-aware inputs, document-level OCR retrieval, clustering, compositionality probing, and a newly designed Visual Semantic Textual Similarity (Visual STS) category. Tasks are sourced from a broad set of public datasets including multilingual retrieval benchmarks (xFlickr&CO, XM3600 [26], WIT [4]), document understanding corpora [11], compositionality tests (ARO [29], WinoGround [27], SugarCrepe [13]), and standard vision classification sets [8].

A key novelty of MIEB is **Visual STS**: we render standard STS text pairs (STS12–17 and STS-b) as images and evaluate encoders on their ability to assign semantically appropriate similarity scores purely from visual content. This probes visual text understanding in a way that standard retrieval benchmarks cannot. We show that Visual STS scores are strongly correlated ($>$99%) with performance on OCR and chart understanding tasks in generative MLLM benchmarks [28], making it an efficient proxy for encoder quality in downstream MLLM applications.

To make evaluation practical, we introduce **MIEB-lite**, a 51-task subset derived via task-correlation analysis (UMAP [21] + HDBSCAN [20]) that achieves $\sim$82% runtime reduction while maintaining a Spearman correlation of 0.992 with the full benchmark ranking. Evaluation is unified through the MTEB codebase and hosted on a public leaderboard.

Our main contributions are:

- **MIEB**: a unified benchmark of 130 tasks across 8 categories and 38 languages for image and image-text embeddings with a public leaderboard.
- **Visual STS**: a novel evaluation protocol that renders text pairs as images to diagnose visual text understanding, and demonstrates strong correlation with MLLM perfor-

mance.

- **MIEB-lite**: an efficient 51-task subset derived via task-correlation clustering with $\sim 82\%$ runtime savings and minimal diagnostic loss.
- **Empirical analysis** of 20+ models, revealing that no single model dominates and identifying complementary strengths across encoder families.

2. The MIEB Benchmark

2.1. Design Principles

MIEB is built around four principles. *Comprehensiveness*: tasks span the full breadth of real-world image embedding use cases, from fine-grained retrieval to abstract compositionality. *Reproducibility*: all models are evaluated in zero-shot mode (frozen weights) except for lightweight linear probes trained on frozen embeddings with 16 shots per class; the MTEB codebase enforces standardized preprocessing and metric computation. *Multilinguality*: 38 languages are represented to stress under-evaluated multilingual generalization. *Efficiency*: MIEB-lite enables rapid prototyping without sacrificing ranking fidelity.

2.2. Task Categories

Table 1 provides an overview of all 8 task categories.

Retrieval (45 tasks) is the largest category, covering image–image, image–text, text–image, and interleaved query/document combinations across 38 languages. It includes standard cross-modal benchmarks (e.g., MSCOCO [19]) as well as multilingual variants (xFlickr&CO, XM3600 [26], WIT [4]) and instruction-aware retrieval settings. The metric is nDCG@10.

Document Understanding (10 tasks) uses the retrieval framework to assess OCR and layout-understanding capabilities, building on the ViDoRe benchmark [11]. Queries and documents mix text and image modalities. The metric is nDCG@5.

Linear Probing (22 tasks) trains a logistic regression classifier (max 100 iterations) on frozen embeddings with 16 shots per class, covering datasets such as ImageNet-1K [8], CIFAR-10/100 [18], OxfordPets, Flowers102, Food101 [3], and StanfordCars. Accuracy is the metric.

Zero-shot Classification (23 tasks) maps image embeddings to nearest text-label embeddings ("a photo of [class]") without any training. Standard vision benchmarks are used [16, 17]. Accuracy is the metric.

Clustering (5 tasks) applies k-means (with k equal to the number of ground-truth labels) to image embeddings and measures quality with Normalized Mutual Information (NMI) [7].

Compositionality (7 tasks) evaluates sensitivity to linguistic and visual confounders via pair-classification tasks. Datasets include ARO [29], WinoGround [27], and Sugar-Crepe [13]. Accuracy is the metric.

Vision-centric QA (6 tasks) tests counting and object-detection understanding via retrieval-style tasks where interleaved (image + question) queries are matched to images or text. Tasks span scene-level object counting and spatial-relation probing, where each query pairs a natural-language question with an image and the model must retrieve the correct answer from a gallery of images or text responses. Accuracy/nDCG is used.

Visual STS (9 tasks) is our novel category. We render the text pairs from STS12–17 and STS-b [2, 5] as images using a fixed font and layout, then compute cosine similarity between image embeddings and measure Spearman correlation with human similarity ratings. This probes whether an encoder can read and understand text as an image—a capability needed for document understanding and OCR-heavy retrieval. The rendering pipeline uses a fixed sans-serif font at a consistent resolution (224px square), ensuring that all models face a comparable visual decoding challenge regardless of their native input resolution.

2.3. Modality Handling

Many tasks involve both image and text modalities simultaneously (interleaved inputs). For models that natively support interleaved inputs (e.g., MLLMs such as E5-V [14] and VLM2Vec [15]), we pass the combined input directly. For models lacking native interleaving support (e.g., CLIP [23] and SigLIP [30]), we sum the image and text embeddings to form a single representation. This approach allows a fair, unified evaluation framework across fundamentally different architectures.

2.4. MIEB-lite

Running the full 130-task benchmark can be prohibitively expensive, especially for large MLLM-based encoders (e.g., 264 GPU-hours for E5-V on an NVIDIA H100). We derive MIEB-lite by analyzing pairwise task-performance correlations across all evaluated models. Using UMAP [21] for dimensionality reduction and HDBSCAN [20] for clustering, we identify 51 representative tasks that cover all 8 categories while minimizing redundancy. Table 2 shows that MIEB-lite reduces runtime by $\sim 82.4\%$ for an 8B-parameter model (46.4 vs. 264.0 GPU-hours) and by $\sim 72.9\%$ for a lightweight CLIP model. The Spearman rank correlation between MIEB-lite and full MIEB rankings is 0.992, confirming that the subset preserves the diagnostic power of the full benchmark.

3. Models

We evaluate more than 20 models covering three broad families:

Task category	Example abilities assessed	# Tasks	# Languages	Modalities
Retrieval	cross-modal/-lingual matching	45	38	i-i; i-t; t-i; it-i; it-t; i-it; t-it; it-it; i-t
Document Understanding (Retrieval)	OCR abilities	10	2	t-i; i-t; it-t
Linear Probing (Classification)	information encoded	22	1	i-i; i-i
Clustering	embedding space consistency	5	1	i-i
Zero-shot Classification	cross-modal matching	23	1	i-t; i-t
Compositionality Evaluation (PairClassification)	reasoning with confounders	7	1	i-t; t-i
Vision-centric QA (Retrieval)	counting, object detection	6	1	it-t; it-i
Visual STS	OCR abilities	9	12	i-i
MIEB	all	130	38	all
MIEB-lite	all	51	38	all

Table 1. **An overview of MIEB tasks.** In brackets behind task categories, we denote the task type implementation in the code, e.g., our document understanding tasks use our retrieval implementation. We denote the modalities involved in both sides of the evaluation (e.g., queries and documents in retrieval; images and labels in zero-shot classification) with i=image, t=text.

Model Name	# Params (M)	Runtime (NVIDIA H100 GPU hours)		
		MIEB	MIEB-lite	Reduction %
E5-V	8360	264.0	46.4	82.4% ↓
CLIP (base-patch32)	151	16.6	4.5	72.9% ↓

Table 2. **MIEB vs. MIEB-lite runtime comparison.**

Contrastive Encoders (Enc.) are dual-encoder models trained with image-text contrastive objectives. We evaluate multiple variants of CLIP [23] (ViT-B, L, H, g, bigG backbones; trained on LAION-2B [6], DataComp [12]), SigLIP [30] (base/large/so400m; resolutions 224–512), and EVA02-CLIP [24] (bigE backbone).

MLLM-based Embedders (MLLM) extract embeddings from the hidden states of a large multimodal language model, often via special pooling tokens. We evaluate:

- **E5-V** [14]: a universal MLLM embedder fine-tuned for image-text embedding tasks that achieves strong text STS scores.
- **VLM2Vec** [15]: a multimodal MLLM embedding model trained for diverse multimodal tasks; evaluated in LoRA and full fine-tuning variants.
- **Voyage-multimodal-3** [1]: a commercial MLLM-based embedding model supporting interleaved text, images, and screenshots.

Additionally, we include VISTA [31] (a model integrating visual and text embeddings for universal retrieval) in our multilingual analysis.

4. Implementation Details

All models are evaluated in **zero-shot** mode with frozen weights, except for linear probing tasks, which train a logistic regression classifier on frozen image embeddings using 16 shots per class. To assess the impact of training set size, we also evaluate performance with $k \in \{1, 2, 4, 8, 16, 32, 64\}$ shots (see Figure 2). For zero-shot classification, prompts follow the standard template "a photo of a [class]".

Retrieval and Document Understanding use the standard MTEB retrieval pipeline with nDCG@10 (nDCG@5 for document tasks). For clustering, k-means is run with k set to the number of ground-truth label classes, and NMI is computed. For Visual STS, Spearman correlation between cosine similarities and human annotations is used.

Interleaved inputs: For models lacking native interleaving support, image and text embeddings are summed element-wise. MLLM-based models receive the interleaved sequence as a single input.

Evaluation infrastructure is built on the MTEB codebase [22], with strict dataset and model versioning to ensure reproducibility. Results are hosted on the MTEB public leaderboard (`https://hf.co/spaces/mteb/leaderboard`).

5. Experimental Results

5.1. Overall MIEB Results

Table 3 shows results for the top 20 models on MIEB (130 tasks) and MIEB-lite (51 tasks). The key findings are:

No single model dominates. Voyage-multimodal-3 achieves the highest overall mean on MIEB (62.5) and MIEB-lite (58.1), but different model families lead on different categories. EVA02-CLIP-bigE-14-plus leads clustering (92.4 NMI) and linear probing (86.0 accuracy), while CLIP-ViT-bigG-14 achieves the best English retrieval (41.5 nDCG@10). VLM2Vec variants lead Vision-centric QA (62.0–62.1).

MLLM embedders excel on OCR and multilingual tasks. E5-V achieves 66.6 on multilingual retrieval (mean across 38 languages) and 79.3 on Visual STS, while Voyage-multimodal-3 leads document understanding (71.1 nDCG@5) and multilingual Visual STS (70.4). Standard contrastive encoders score substantially lower on these tasks.

Contrastive encoders are stronger on classification

tasks. SigLIP-so400m-patch14-384 achieves 84.6 on linear probing and 70.8 on zero-shot classification, comparable to EVA02-CLIP (86.0 LP, 70.8 ZS), both substantially outperforming MLLM-based models on these categories.

MIEB-lite preserves rankings. The Spearman correlation between MIEB and MIEB-lite rankings is 0.992, confirming that the 51-task subset is a reliable proxy for the full benchmark.

5.2. Multilingual Retrieval

Table 4 presents performance on the three multilingual retrieval benchmarks (xFlickr&CO, XM3600, WIT) across 38 languages.

E5-V dominates multilingual retrieval with an average of 74.3 nDCG@10 across datasets, compared to 66.8 for SigLIP and 54.4 for VISTA. Notably, E5-V also shows the lowest per-language variance (1.4) among the best models, indicating more consistent cross-lingual generalization. Standard contrastive encoders (Open-CLIP, EVA02-CLIP) perform substantially worse (31.0–31.4 average), highlighting that contrastive training on English-dominated data does not generalize well to multilingual retrieval.

5.3. Visual STS: Probing Visual Text Understanding

Table 5 compares E5-V performance on standard (text) STS benchmarks [2, 5] and our Visual STS. E5-V achieves a mean of 86.0 on text STS versus 79.3 on Visual STS, a gap of 6.7 points. This difference quantifies how much information is lost when the model must read text from an image rather than process it as tokens—near-complete but not perfect visual text understanding.

Figure 1 shows that Visual STS performance is strongly correlated (>99%) with performance on generative MLLM benchmarks (OCR, chart understanding) from Cambrian [28]. This makes Visual STS a practical diagnostic for selecting vision encoders for MLLM systems: a simple image-only evaluation predicts downstream MLLM performance without requiring full model training.

5.4. Linear Probing: Effect of Shot Count

Figure 2 shows how linear probing accuracy scales with the number of shots k per class. Performance increases rapidly from $k = 1$ to $k = 16$ and begins to plateau around $k = 32$–64. We adopt $k = 16$ as the standard in MIEB, offering a practical balance between evaluation cost and informativeness. CLIP and SigLIP models consistently outperform MLLM-based embedders at all shot levels on fine-grained classification tasks, confirming that contrastive training on large image-text datasets produces superior low-shot visual classifiers.

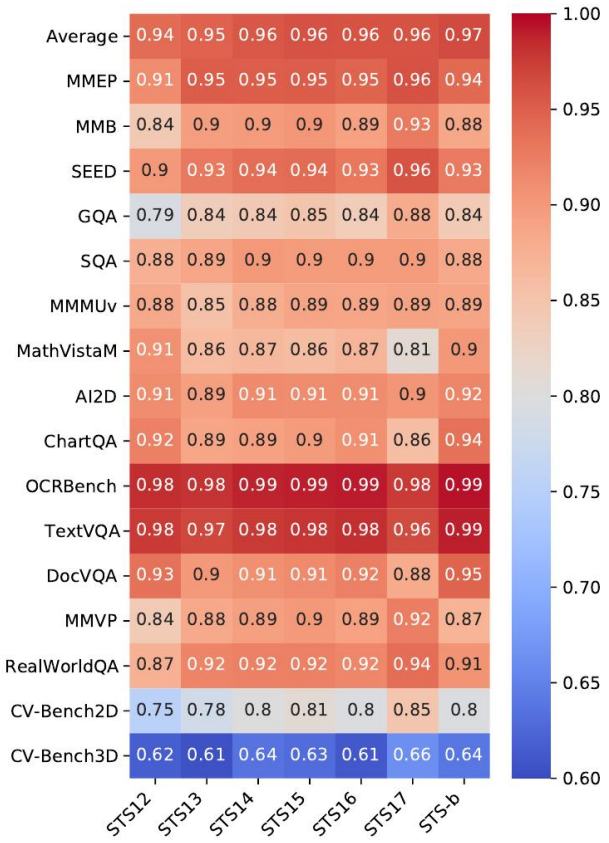

Figure 1. **Correlations between performance on generative MLLM benchmarks from Tong et al. [28] (y-axis) and our Visual STS (x-axis).** Visual STS strongly predicts MLLM performance on OCR and chart tasks, with correlations exceeding 99%.

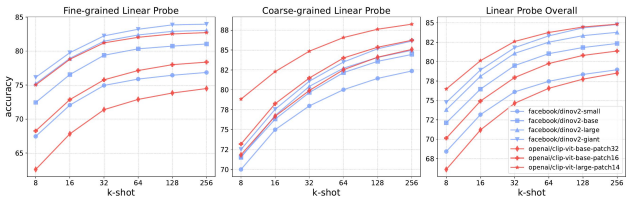

Figure 2. **Linear probing performance across different shots k.** Performance increases quickly up to $k = 16$ and plateaus, motivating our choice of 16-shot as the default.

5.5. Zero-shot Classification and Compositionality

Zero-shot classification reveals a clear hierarchy: EVA02-CLIP-bigE-14-plus and SigLIP-so400m-patch14-384 both achieve 70.8 accuracy, benefiting from large-scale contrastive pretraining on diverse image-text pairs. MLLM-based embedders trail by 10–15 points on average, confirming that large language model backbones do not inherently yield stronger zero-shot visual classifiers. On compositionality tasks, however, the gap between model families nar-

MIEB Full (130 tasks)													
Model Name (↓)	**Model Type**	**Rtrv. (45)**	**Clus. (5)**	**ZS. (23)**	**LP. (22)**	**Cmp. (7)**	**VC. (6)**	**Doc. (10)**	**vSTS (en) (7)**	**Rtrv. (m) (3 (55))**	**vSTS (x&m) (2 (19))**	**Mean (en) (125)**	**Mean (m) (130)**
Voyage-multimodal-3	MLLM	38.8	82.4	58.2	71.3	43.5	48.6	**71.1**	**81.8**	58.9	**70.4**	**62.0**	**62.5**
E5-V	MLLM	34.0	70.0	50.0	74.5	**46.3**	51.9	62.7	79.3	**66.6**	46.3	58.6	58.2
siglip-so400m-patch14-384	Enc.	40.8	82.1	**70.8**	84.6	40.4	46.3	56.4	68.0	40.2	41.4	61.2	57.1
siglip-large-patch16-384	Enc.	39.9	79.9	68.0	83.7	39.7	45.4	53.3	69.5	51.1	39.8	59.9	57.0
siglip-large-patch16-256	Enc.	38.8	82.1	67.7	82.5	40.8	44.9	39.4	67.4	49.8	38.1	57.9	55.2
siglip-base-patch16-512	Enc.	38.1	74.7	64.1	80.9	37.5	53.2	52.1	67.7	43.2	38.1	58.5	54.9
CLIP-ViT-bigG-14-laion2B	Enc.	**41.5**	85.6	69.4	83.6	42.4	43.2	43.2	70.9	28.0	34.5	60.0	54.2
siglip-base-patch16-384	Enc.	37.7	76.3	64.1	80.6	38.5	52.8	45.0	67.0	42.5	37.5	57.8	54.2
EVA02-CLIP-bigE-14-plus	Enc.	40.1	**92.4**	70.8	**86.0**	45.7	39.4	32.3	72.0	27.8	28.2	59.8	53.5
CLIP-ViT-L-14-DataComp.XL	Enc.	38.1	86.4	68.4	82.0	39.1	52.3	38.6	69.9	23.8	35.8	59.4	53.4
siglip-base-patch16-256(m)	Enc.	35.6	74.6	61.2	78.9	38.1	51.3	26.4	65.5	59.2	40.3	53.9	53.1
CLIP-ViT-H-14-laion2B	Enc.	39.7	83.9	67.5	82.5	42.0	45.8	40.4	65.5	25.5	33.9	58.4	52.7
CLIP-ViT-g-14-laion2B	Enc.	39.8	82.7	67.9	82.8	41.9	44.2	37.6	69.1	25.9	31.7	58.3	52.4
EVA02-CLIP-bigE-14	Enc.	39.0	89.4	69.3	84.5	42.4	43.6	31.6	68.8	25.5	28.3	58.6	52.2
siglip-base-patch16-256	Enc.	36.6	75.2	63.1	79.7	39.5	52.2	31.7	66.2	41.3	34.4	55.5	52.0
siglip-base-patch16-224	Enc.	36.3	74.5	62.6	79.3	39.8	51.1	26.2	64.3	41.2	33.5	54.3	50.9
CLIP-ViT-L-14-laion2B	Enc.	38.0	83.5	65.8	81.2	40.8	45.9	36.3	65.8	23.0	26.0	57.2	50.6
VLM2Vec-LoRA	MLLM	27.7	72.6	46.3	62.0	34.6	62.0	49.7	72.6	34.9	42.2	53.4	50.5
VLM2Vec-Full	MLLM	27.6	70.7	46.3	62.0	35.4	**62.1**	49.8	72.6	35.0	42.2	53.3	50.4
clip-vit-large-patch14	Enc.	33.7	76.4	62.1	80.1	44.8	44.1	38.0	64.5	20.2	35.1	55.4	49.9
MIEB-lite (51 tasks)													
Model Name (↓)	**Model Type**	**Rtrv. (11)**	**Clus. (2)**	**ZS. (7)**	**LP. (8)**	**Cmp. (6)**	**VC. (5)**	**Doc. (6)**	**vSTS (en) (2)**	**Rtrv. (m) (2 (47))**	**vSTS (x&m) (2 (19))**	**Mean (en) (47)**	**Mean (m) (51)**
Voyage-multimodal-3	MLLM	33.2	76.6	48.6	69.3	35.8	50.0	**63.5**	**84.2**	49.0	**70.4**	**57.7**	**58.1**
siglip-so400m-patch14-384	Enc.	32.4	75.9	73.8	78.8	32.8	48.0	46.9	69.6	35.4	41.4	57.3	53.5
siglip-large-patch16-384	Enc.	31.9	75.2	71.3	77.7	32.1	46.8	44.9	69.6	43.5	39.8	56.2	53.3
E5-V	MLLM	26.9	51.7	36.2	70.6	**39.4**	52.6	56.0	81.2	**58.3**	46.3	51.8	51.9
siglip-large-patch16-256	Enc.	31.0	76.5	70.3	76.3	33.4	46.5	31.9	67.6	42.6	38.1	54.2	51.4
CLIP-ViT-bigG-14-laion2B	Enc.	34.2	80.8	72.4	77.8	35.0	43.0	35.5	73.4	26.2	34.5	56.5	51.3
siglip-base-patch16-512	Enc.	30.8	69.7	66.3	74.6	29.7	55.5	42.6	67.1	34.8	38.1	54.5	50.9
EVA02-CLIP-bigE-14-plus	Enc.	**35.2**	**87.3**	**74.0**	**80.0**	38.9	38.8	26.2	73.7	26.0	28.2	56.8	50.8
siglip-base-patch16-384	Enc.	30.6	72.2	66.0	74.4	31.0	55.1	37.1	66.9	34.5	37.5	54.1	50.5
CLIP-ViT-L-14-DataComp.XL	Enc.	31.0	80.4	69.4	75.3	31.6	54.9	30.8	72.5	22.6	35.8	55.7	50.4
CLIP-ViT-H-14-laion2B	Enc.	32.8	79.3	69.4	76.8	34.8	46.8	33.7	68.3	23.9	33.9	55.2	50.0
EVA02-CLIP-bigE-14	Enc.	34.3	86.7	73.0	78.3	35.1	44.4	25.1	69.9	23.9	28.3	55.9	49.9
siglip-base-patch16-256(m)	Enc.	28.2	68.2	63.2	73.4	30.7	53.3	22.9	63.7	52.9	40.3	50.4	49.7
CLIP-ViT-g-14-laion2B	Enc.	33.5	76.8	69.6	77.3	34.7	45.0	29.9	71.6	24.2	31.7	54.8	49.4
siglip-base-patch16-256	Enc.	29.5	69.6	65.6	73.6	32.2	54.4	25.0	66.1	33.5	34.4	52.0	48.4
CLIP-ViT-L-14-laion2B	Enc.	31.1	76.4	67.8	75.9	33.6	46.9	28.7	68.7	21.4	26.0	53.6	47.6
clip-vit-large-patch14	Enc.	26.7	71.3	63.8	74.5	**39.4**	44.9	29.4	69.4	19.8	35.1	52.4	47.4
siglip-base-patch16-224	Enc.	29.3	68.4	65.0	73.5	32.5	53.0	20.9	64.2	33.6	33.5	50.8	47.4
CLIP-ViT-B-16-DataComp.XL	Enc.	28.3	73.6	61.9	73.2	31.4	56.9	22.7	69.7	19.9	28.5	52.2	46.6
VLM2Vec-LoRA	MLLM	21.0	66.3	32.1	64.8	29.4	**65.3**	42.7	70.9	24.8	42.2	49.1	46.0

Table 3. **MIEB results broken down by task categories for the top 20 models.** We provide averages of both English and multilingual tasks. Models are ranked by the Mean (m) column. Shortcuts are x=Crosslingual, m=Multilingual, en=English, and task categories from Figure 4. We refer to the leaderboard for the latest version: https://hf.co/spaces/mteb/leaderboard

Model Name	xFlickr&CO		XM3600		WIT		avg.	
	avg.	var.	avg.	var.	avg.	var.	avg.	var.
E5-V	**90.8**	**0.1**	**74.8**	3.5	**57.3**	0.6	**74.3**	1.4
SigLIP	80.4	1.2	65.6	5.3	54.4	1.3	66.8	2.6
VISTA (m3)	65.3	0.2	48.5	**2.0**	49.3	**0.4**	54.4	**0.9**
VLM2Vec	63.8	3.8	27.0	4.7	31.7	2.5	40.8	3.6
Open-CLIP	35.9	9.3	20.5	6.0	37.8	6.5	31.4	7.3
EVA02-CLIP	35.6	9.4	20.1	6.0	37.4	6.4	31.0	7.2

Table 4. **Performance of models on multilingual retrieval tasks across 38 languages.** We compute the average performance across languages (avg) and the respective variance (var). We take the best variant from each top-6 model family.

	12	13	14	15	16	17	b	avg.
STS*	80.0	89.9	85.7	89.1	85.9	87.9	83.5	86.0
v-STS (ours)	73.2	78.2	74.9	84.2	79.5	85.8	79.4	79.3

Table 5. **E5-V performance on regular STS and our Visual STS.** *: numbers from Jiang et al. [14]. Columns are STS12-17 and STS-b.

rows considerably: E5-V leads with 46.3 while EVA02-CLIP-bigE-14-plus follows at 45.7, a margin of less than one point. At the dataset level, SugarCrepe [13] is the most

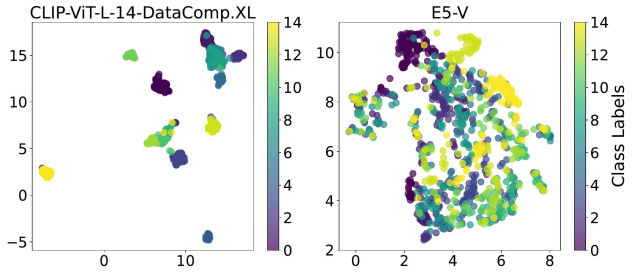

Figure 3. **UMAP Visualization of ImageNet Dog15.** Clusters correspond to 15 fine-grained dog breeds; compact, well-separated clusters indicate a geometrically meaningful embedding space.

tractable compositionality task (most models exceed 70), whereas WinoGround [27] remains extremely challenging (best model score below 30), highlighting how far current embeddings fall short of human-level compositional reasoning regardless of architecture.

5.6. Clustering Visualization

Across the 5 clustering tasks, EVA02-CLIP-bigE-14-plus achieves the highest mean NMI (92.4), confirming that large-scale contrastive pretraining produces geometrically well-structured embeddings for fine-grained visual categories. SigLIP variants follow closely (NMI 89–91), while MLLM-based embedders score noticeably lower (NMI 75–82), consistent with the observation that MLLM encoders are optimized for cross-modal semantic alignment rather than compactness of within-class visual clusters. The performance gap between large and small contrastive encoders (EVA02-bigE: 92.4 vs. CLIP-ViT-B/32: 78.3) indicates that backbone scale contributes meaningfully to cluster separation even on tasks that do not require language understanding.

Figure 3 shows a UMAP visualization of ImageNet Dog15 embeddings from a representative encoder. Well-separated, compact clusters indicate that the embedding space correctly encodes fine-grained visual distinctions between dog breeds. This qualitative view complements the NMI metric in the benchmark table, illustrating how clustering scores translate to embedding space geometry.

6. Discussions

6.1. Complementary Strengths of Model Families

MIEB reveals a clear division of labor among model families. Contrastive encoders (CLIP, SigLIP, EVA02-CLIP) are trained to align image and text representations at scale, producing embeddings that are highly effective for zero-shot classification and linear probing. In contrast, MLLM-based embedders leverage the reasoning and language-understanding capabilities of large language model backbones, producing embeddings that generalize better across languages and to OCR-heavy tasks.

This complementarity suggests that a practical system might benefit from specialized encoders for different subtasks, and MIEB provides the granular task-level scores needed to make such design decisions.

6.2. Challenges in Compositionality and Interleaved Retrieval

Compositionality remains challenging across all model families. WinoGround [27] and ARO [29] probe fine-grained linguistic structure (word order, attribute binding), and even the best models score modestly. E5-V leads compositionality with 46.3, only marginally ahead of EVA02-CLIP-bigE-14-plus (45.7). This suggests that current training objectives—whether contrastive or generative—do not sufficiently stress compositional understanding. Hard negative mining strategies that specifically target compositional contrast—such as swapping attributes or spatial relations in image captions during training—represent a promising direction for future work. The narrow margin separating contrastive from MLLM-based models on these tasks further indicates that simply scaling model size or pretraining data volume is unlikely to close this capability gap on its own.

Interleaved retrieval (queries combining images and text) is similarly challenging, particularly for contrastive encoders that approximate interleaving by summing embeddings. Voyage-multimodal-3, which natively supports interleaved inputs, achieves the best overall score, suggesting that native interleaving support is important for this class of tasks.

6.3. Visual STS as a Diagnostic

The strong correlation between Visual STS and MLLM OCR/chart benchmarks (Figure 1) makes Visual STS an efficient proxy. At just 9 tasks with image-only inputs, it can be evaluated quickly and does not require any text encoder. This is practically valuable: when selecting a vision encoder for an MLLM, a researcher can run Visual STS first as a cheap filter before committing to full-scale evaluation.

6.4. Practical Recommendations

Based on our experimental findings, we offer practical guidance for practitioners selecting image embedding models. For applications requiring multilingual support or heavy OCR and document understanding, MLLM-based embedders such as E5-V or Voyage-multimodal-3 are the preferred choice despite their higher computational cost. For classification-intensive workloads with limited labeled data, SigLIP and EVA02-CLIP variants deliver superior low-shot accuracy at a fraction of the inference cost. When interleaved image-text queries are central to the application, native interleaving support (as in Voyage-multimodal-3) out-

performs the approximation of summing separate embeddings. Finally, MIEB-lite provides a cost-effective screening step: practitioners can rapidly benchmark candidate models on the 51-task subset before committing to full evaluation, saving up to 82% in compute while preserving ranking fidelity.

6.5. Model Scale and Efficiency Trade-offs

An important observation from MIEB is that model scale and task performance are not uniformly correlated. EVA02-CLIP-bigE-14-plus achieves leading classification and clustering scores despite being a single-modality contrastive encoder, while MLLM-based embedders—which carry substantially larger parameter counts—outperform contrastive encoders primarily on tasks that stress language understanding (multilingual retrieval, Visual STS, document understanding) rather than pure visual recognition. Smaller CLIP-ViT-B variants remain surprisingly competitive on zero-shot classification relative to their size. This pattern suggests that practitioners should rely on MIEB's category-level scores for task-specific model selection rather than overall averages, particularly when deploying under real-world compute constraints. The MIEB-lite runtime analysis makes this trade-off concrete: the 8B-parameter VLM2Vec requires 264 GPU-hours for a full benchmark run compared to under 20 GPU-hours for a ViT-B CLIP model, underscoring the value of efficient subset evaluation when comparing architectures across scales.

6.6. Limitations

MIEB currently focuses on task-average metrics and does not account for dataset difficulty differences across tasks within a category. The leaderboard mean weights all tasks equally, which may slightly favor models trained on data distributions similar to popular benchmarks. Future work could introduce difficulty-adjusted or query-distribution-aware aggregations. Additionally, the 16-shot linear probing protocol may not fully reflect deployments with larger labeled sets or continual fine-tuning; investigating how MIEB scores correlate with such settings is an important open problem. MIEB does not currently include video or 3D modalities; extending to these domains is a further valuable direction.

Furthermore, the Visual STS rendering pipeline, though standardized, may introduce artifacts—such as font antialiasing differences or resolution scaling effects—that partially conflate measured performance with preprocessing sensitivity rather than genuine visual text understanding. Future versions of MIEB will investigate typographic variations and rendering conditions. Another avenue for improvement is the aggregation methodology: the current equal-weighted mean across tasks within each category may not reflect the relative importance of individual tasks in real-world deployments. Weighted aggregations informed by application frequency or downstream impact could provide more actionable guidance, and stratified analysis of per-language and per-domain performance within the multilingual retrieval category would offer finer-grained insights into cross-lingual generalization.

7. Related Work

Text Embedding Benchmarks. MTEB [22] unified 56 text-embedding tasks into a single leaderboard, catalyzing rapid progress in sentence embedding research. MMTEB [10] extended this to multilingual settings. BEIR [25] provided a heterogeneous retrieval benchmark across diverse domains. MIEB brings an analogous unification to the vision domain.

Vision-Language Benchmarks. A rich ecosystem of benchmarks covers individual aspects of vision-language understanding: MSCOCO [19] for captioning and retrieval, IGLUE [4] for multilingual transfer, SugarCrepe [13] and ARO [29] and WinoGround [27] for compositionality, ViDoRe [11] for document retrieval, and Cambrian [28] for generative MLLM evaluation. MIEB consolidates 130 such tasks into a single framework.

Image Embedding Models. CLIP [23] established contrastive image-text pretraining at scale. SigLIP [30] replaced the InfoNCE loss with a sigmoid loss, improving sample efficiency. EVA02-CLIP [24] scaled the vision encoder backbone. VISTA [31] integrated visual and text embeddings for universal retrieval. On the MLLM side, E5-V [14] fine-tunes a multimodal LLM for embedding tasks, and VLM2Vec [15] trains on diverse multimodal task mixtures. Voyage-multimodal-3 [1] is a commercial system supporting interleaved inputs. MIEB provides the first unified evaluation across all these model families.

Efficient Benchmarking. Constructing efficient subsets of large benchmarks has been studied in NLP [22]. Our task-correlation-based approach for MIEB-lite is inspired by this line of work but adapted to the visual domain, using UMAP [21] + HDBSCAN [20] clustering of per-task performance profiles.

8. Conclusion

We introduced MIEB, the Massive Image Embedding Benchmark, which unifies 130 evaluation tasks across 8 categories and 38 languages for image and image-text embeddings. MIEB reveals that no single model family dominates all evaluation categories: contrastive encoders excel at classification and clustering, while MLLM-based embedders generalize better to OCR-heavy and multilingual tasks. Our novel Visual STS category—which renders text pairs as images—proves to be a strong predictor of MLLM performance on OCR and chart tasks, providing a cheap diag-

nostic for encoder selection. MIEB-lite, a 51-task efficient subset, achieves $\sim$82% runtime reduction with near-perfect ranking fidelity (Spearman $\rho = 0.992$). By standardizing evaluation through the MTEB codebase and a public leaderboard, MIEB lowers the barrier to rigorous benchmarking and provides the community with the infrastructure to track progress in image embedding research.

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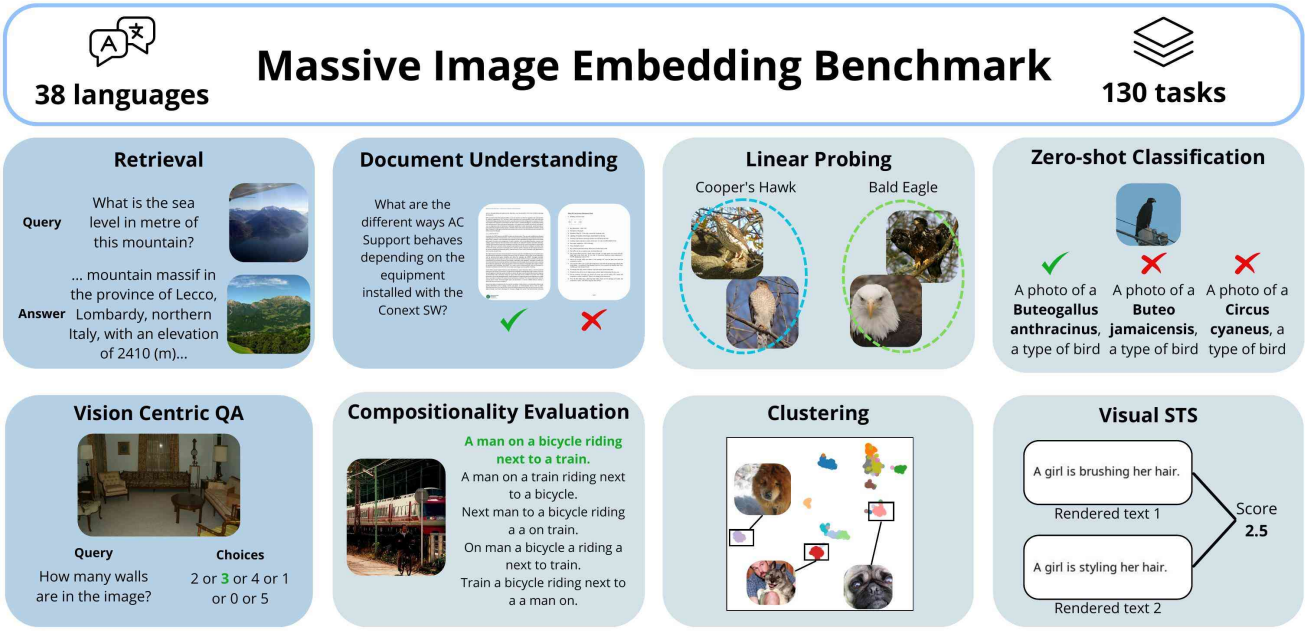

Figure 4. **Overview of MIEB task categories with examples.** MIEB covers 8 task categories spanning retrieval, document understanding, linear probing, zero-shot classification, clustering, compositionality, vision-centric QA, and the novel Visual STS category. Modalities include pure image, pure text, and interleaved image-text inputs.

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